MILITARY SPECIFICATION

TEST SET, INDICATOR-GENERATOR, TACHOMETER TTU-27/E, AIRCRAFT INSTRUMENTS, FIELD

1. SCOPE

1.1 This specification covers one type of aircraft tachometer test set designated Type TTU-27/E.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-315	Aluminum Alloy 52S Bars, Rods, and Wire (Rolled or Drawn)
QQ-A-318	Aluminum Alloy 52S; Plate and Sheet
QQ-S-571	Solder; Lead Alloy, Tin Lead Alloy and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber

Military

MIL-E-1	Electron Tubes and Crystal Rectifiers
JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-B-130	Barrier Material, Paper, Noncorrosive
MIL-B-138	Boxes, Wood, Fiberboard-Lined for Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)
MIL-A-148	Aluminum Foil
MIL-T-945	Test-Equipment, For Use With Electronic Equipment; General Specification
MIL-C-3767	Connectors, Electrical (Power, Bladed Type), General Specification For
MIL-P-4185	Paper, Tissue, Wrapping
MIL-E4682	Electron Tubes, Choice and Application Of
MIL-C-5015	Connectors, Electrical, “AN” Type
MIL-C-5028	Drawings and Data Lists, Preparation Of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-G-8512	Ground Support Equipment, General Requirements For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-M-103O4	Meters, Electrical Indicating, Panel Type, Ruggedized
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-W-16878	Wire, Electrical, Insulated High Temperature
MIL-I-25686	Indicator, Electrical Tachometer ERU-1/A, Crankshaft Speed, 0-4000 RPM, 2 Inch Size.

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-889	Dissimilar Metals
MIL-STD-1653	Power Cable Assemblies
MS3102	Connectors, Electric, Receptacles, Box Mounting
MS3106	Connectors, Electric, Plugs, Straight
MS3507	Connector, Plug, Electric, Solder Contact, Quick Disconnect, Utility
MS28054	Generator, Tachometer, Two-Pole, Aircraft

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Air Force - Navy Aeronautical

AN3027	Switch - One Hole Mounting Double Pole Toggle
AN3057	Adapter - Electrical Accessory to Cable
AN3226	Switch - Three Pole Mounting Three Pole Toggle
AN5547	Generator, Tachometer, Four Pole
AND20005	Drive, Type XV Engine Accessory

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

143	Specifications and Standards; Use of
405	Storage Life - Aeronautical Articles

U.S. Air Force

F29	Codes for Aircraft Finishing Schemes

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3).

3.2 Component parts. The tester shall consist of a single integral unit incorporating a carrying case, a motor driven variable speed drive, a precision rotational speed indicating system, an alternating current voltmeter, load resistors, control switches and test cables.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to ANA Bulletin 143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

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3.3.2 Standard parts. With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Material. Materials shall conform to applicable specifications as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose.

3.4.1 Metals. Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during storage and normal service use.

3.4.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other, Dissimilar metals are defined by MIL-STD-889.

3.4.3 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the acquiring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.4 Protective treatment. When materials are used in the construction of the test set that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The test set shall be designed as a portable unit suitable for testing aircraft tachometer indicators and generators. A precision speed reference (rotational speed indicating system) shall be incorporated for monitoring the output speed of the variable drive. A standard four pole aircraft tachometer generator shall be incorporated within the test set for driving aircraft tachometer indicators under-going test. Provisions shall be incorporated for testing aircraft tachometer generators individually by the resistance load-output voltage method. Provisions shall also be incorporated for permitting individual generators underdoing test to be electrically connected to respective aircraft tachometer indicators. Specification MIL-G-8512 shall be used as a guide in the design of the test set. The construction of the test set shall be mechanically and electrically sound, suitable for the intended purpose, and shall be such as to give permanence in the accuracy of the measurements and stability in the ouput speeds of the variable drive. The design and construction of the test set shall take into account the temperature, humidity, vibration and operational requirements specified herein.

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3.5.1 General layout. The general layout of the test set shall conform substantially to Figures 1 and 2. The detailed mechanical and electrical design of the test set shall be accomplished by the contractor subject to the requirements specified herein, these requirements being detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence of the same. The design, layout and assembly of the test set and its component parts shall be such as to facilitate quantity production and ease of maintenance and operation.

3.6 Performance. The test set shall be capable of meeting the test requirements specified in section 4 when subjected to the following conditions:

a. Temperature. Temperatures ranging from -55°C (-67°F) to +71°C (160°F)

b. Vibration. Vibration at frequencies varying between 10-55 cycles per second at a total excursion of 0.03 + 0.006 inch for a period of 1 hour in each of three mutually perpendicular axes.

c. Endurance. Intermittent operation (15 + 2 minutes on 15 + 2 minutes off) for a period of 48 hours while operating at 4000 + 50 rpm.

d. Humidity. Relative humidity up to 100 percent for a period of 240 hours (10 cycles).

3.7 Standard electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-T-945.

3.7.1 Electron tubes and transistors. Electron tubes shall conform to MIL-E-1. Selection of tubes shall be in accordance with MIL-E-4682. Transistors shall be subject to approval by the acquiring activity.

3.8 Case. The test set shall be enclosed in a carrying case as shown on Figure 1. The storage compartment may be located as shown or on the left side of the panel. The case shall be so designed to facilitate inspection and maintenance of the component parts incorporated therein.

3.8.1 Case material. The case and control panel shall be fabricated from aluminum alloy conforming to QQ-A-318. Material required for reinforcement purposes shall conform to QQ-A-315, as applicable. The thickness of the material employed shall be such as to produce a sturdy case.

3.8.2 Case finish. The inside and outside surfaces of the case, including the panel, shall be anodized in accordance with MIL-A-8625. Th inside and outside of the case shall be finished in final form in accordance with U.S. Air Force Bulletin F29, code F29-8, color 13538 as specified in FED-STD-595. The outside surface of the panel shall be finished in lusterless black.

3.8.3 Dimensions. The outside dimensions of the case, including the projections introducted by the latches, feet and handle when folded down, shall be maintained within a maximum length of 21 inches, a width of 10 inches, and depth of 12 inches.

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3.8.4 Feet. The case shall be provided with 4 feet in the hinged side of the case and 4 feet on the side opposite the lid. The feet shall be formed by dimpling or other suitable means.

3.8.5 Handle. A hinged metal handle shall be provided on the outside of the case. It shall be located as shown in Figure 1 or on top of the lid, and in the optimum position with respect to weight distribution.

3.8.6 Lid. The case shall be provided with a nondetachable lid secured to the case by hinge of the continuous type. Two sturdy lid supports fabricated from corrosive resistance steel shall be provided as shown on Figure 1 for the purpose of relieving hinge strain when the lid is open. Provisions shall be incorporated at the case and cover joint line to maintain the sides of the lid rigid and flush with the respective sides of the case when the lid is closed and the latches secured.

3.8.7 Latches. Sturdy trunk-type pull-down latches shall be provided for securing the transit case as shown on Figure 1. The latches shall be fabricated from corrosive resistance steel.

3.8.8 Accessory storage, A compartment shall be provided in the case as shown on Figure 1 for storage of adapter cables and the power line connector cable.

3.8.9 Ventilation. The case shall be designed so as to be capable of providing adequate ventilation for the equipment mounted below the control panel.

3.8.10 Mounting brackets. Rigidly constructed mounting brackets shall be provided for mounting and securing the variable speed drive and motor.

3.8.11 Control panel. The case shall incorporate a control panel as shown on Figure 1 for mounting the rotational speed indicator, the variable speed drive and control, the alternating current voltmeter, the tachometer generator mounting pad, control switches and electrical receptacles. The panel layout shall be substantially as shown on Figure 2.

3.8.11.1 Markings. The test and approximate locations of markings for the control panel shall be substantially as shown on Figure 2. The markings shall be engraved in the panel and shall be 0.125 + 0.010 inch in height and 0.020 + 0.005 inch in width. The depth of the markings shall be 0.015 + 0.005 inch. The markings shall be filled with enamel or other suitable material, white in color. In lieu of engraving, the wrinlay wrinkle process, or equal maybe used.

3.9 Controls. All controls provided for the test set shall be readily accessible, suitably arranged, and of such size and construction to permit convenience and ease of operation under all applicable service conditions by operating personnel. The controls shall rotate freely and smoothly without binding or excessive lost motion. The controls shall be adequately lubricated, where necessary, and such lubrications shall be selected and applied in a manner that the test set will meet the requirements specified herein. The control knobs shall be firmly secured to their respective control shafts by suitable retaining devices.

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3.9.1 Control knobs. Each rotary-type switch shown on Figure 2 shall be furnished with a one-piece molded General Radio Type KNB-1 knob, or equal, black in color, and provided with a bored brass insert and bushing for a 1/4-inch shaft utilizing two set screw 90° apart. The knob shall incorporate a bar type extension to facilitate switching operations. The pointer shall be provided with a white reference marking. The control knob provided for controlling the variable speed drive shall be of suitable size, configuration and design to permit rapid and precise setting of the output speed of the test set by operating personnel over the complete design range. The knob shall incorporate an auxiliary spinner.

3.10 Variable speed drive. The variable speed drive shall consist of a electric motor, associated gearing, controls, and a means for providing signals to drive the precision rotational speed indicator.

3.10.1 Variable speed drive control. The drive shall be designed to be infinitely variable in output speed over the complete range from O to 5000 revolutions per minute. The design shall be capable of adequately transmitting the horsepower produced by the motor and as dictated by the overall load requirements of the test set. The torque required to change the output speed of the drive during operation by means of the control shaft shall not be less than 3 nor more than 6 inch-ounces. When the test set is not in operation, the drive shall not be damaged in the event the control knob is rotated. The design shall be trouble free and suitable for continuous duty usage. It shall be quiet running and capable of providing a high degree of stability.

3.10.2 Motor. The electric motor required for driving the variable speed drive shall be designed to operate directly from a single phase, 400 + 20 cycle, 115 + volt A-C power source or through a metallic rectifier power unit. The motor shall be suitable for continuous duty usage. It shall be capable of producing the horsepower required by the tester for fulfilling the tests specified in section 4. During accomplishment of applicable tests, the temperature rise of the motor shall be maintained within a limit which shall not adversely affect the performance characteristics of the test set. It shall be equipped with an enclosure capable of adequately protecting the operation parts and windings. The enclosure shall be designed to mount on the housing provided for variable speed transmission. The bearings shall be of the ball-bearing type and shall be grease-packed.

3.10.2.1 Power supply switch. The electric motor shall be electrically energized and de-energized by means of a double pole toggle switch conforming to AN3027-9. Overload protection shall be provided between the input electrical connector receptacle and the power switch.

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3.10.3 Gear box. The gear box shall be enclosed within a rigidly constructed housing equipped with two aircraft generator mounting pads. One of the pads shall be located at the level of the control panel as shown on Figure 2 to provide a drive outlet for testing tachometer generators. The second pad shall be located inside the case and shall have an AN5547-2 tachometer generator attached thereto. The gear box shall be designed for continous duty at any speed over the range from O to 5000 revolutions per minute when connected to the output load of the test set. The gearing shall be adequately lubricated, The drives in the tachometer generator mounting pads shall rotate simultaneously in the clockwise direction (when viewed from the outer mounting end of the pads) and at the same speed. The mounting pads shall conform to AND20005 insofar as mounting arrangement and basic drive connection details are concerned. With respect to the pad mounted on the panel, provisions shall be incorporated for permitting tachometer generators to be quickly and easily attached and detached therefrom. An additional pad may be mounted on the panel to facilitate testing of 4-pole generators.

NOTE: The requirements outlined in the foregoing paragraph shall not be construed as meaning that the master AN5547-2 four pole generator incorporated within the test set will be driven at a speed in excess of 2500 rpm under load.

3.11 Rotational speed indicating system. The rotational speed indicating system (digital speed indicator) shall represent a stable and precision remote indicating system of the electronic type, It shall be designed to accept electrical output signals from either the master 4-pole generator contained within the test set or from any external 2- or 4-pole standard tachometer generator, and to convert these signals into indications of rpm x1, rpm x2, or percent rpm, consistent with the setting of the selector switch. The speed registered by the indicating system shall be directly related and based upon the speed of the output drives incorporated within the tachometer mounting pads.

NOTE: Tachometer generators of the four pole type are designed to be driven at one half engine speed and produce a three phase output voltage having a frequency of two cycles, resulting in the associated tachometer indicator registering full engine speed. Tachometer generators of the two pole type are driven at full engine speed and produce a three phase output voltage having a frequency of one cycle, resulting in the associated tachometer indicator registering full engine speed. Tachometer generators of the four pole type are normally furnished for reciprocating engine aircraft with the indicators calibrated in revolutions per minute. Tachometer generators of the two pole type are normally furnished on jet engine aircraft with the indicators calibrated in percent rpm.

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3.11.1 Electronic components. The rotational speed indicating system shall incorporate a minimum number of electronic components including vacuum tubes. All the components shall be operated at conservative levels. The circuits shall be so designed that the aging of the vacuum tubes will not adversely affect the accuracy of the indicating system. The system shall be capable of monitoring the ouput speed of the variable drive and of increasing or decreasing automatically and consecutively with like changes in the output speed of the variable drive. Transistors shall be employed in the circuits in lieu of vacuum tubes wherever practicable.

3.11.2 Display. An in-line type display consisting of not more than 5 nor less than 4 digits shall be provided as shown on Figure 2. The display shall be designed to indicate in rpm multiplied by 1, rpm multiplied by 2 and percent rpm depending upon the setting of the "RPM-PERCENT RPM" switch located on the panel. In lieu of the selector switch approach specified, the display may incorporate individual digital indications of percent rpm, rpm x 1, and rpm x 2 which function simultaneously and do not require a selector switch. Ranges from O to 5,000, 0 to 10,000 (9999) and O to 119.04 (119.0) shall apply to the indications in rpm x 1, rpm x 2, and percent rpm, as applicable to the 4- or 5- digit displays. When the indicator is set to indicate in percent rpm a decimal point shall register between the third and fourth digits reading from left to right. Unless otherwise approved, the display shall incorporate neon gas discharge type tubes or drum type counter wheels. The basic design selected shall be capable of displaying digits 0-9 in any selected order as controlled by the associated switching circuits. The basic size of the numerals shall not be less than 0.187 inch or more than 0.5 inch in height. Electrically energized numerals, if employed, shall provide a satisfactory degree of readability under sunlight conditions.

3.11.3 Fuse. The rotational speed indicating system shall be designed to operate from a 115 + 10V 400 + 20 cycle, single-phase power source. The input power circuit to the indicating system shall be protected with a suitable fuse which shall be contained within a holder mounted on the panel. The fuse shall be replaceable from the front of the panel.

3.11.4 Accuracy. The rotational speed indicating system shall be capable of providing measurement accuracies of + 10.0 rpm, + 20.0 rpm, and + 0.2 percent rpm respectively for the rpm x 1, rpm x 2, and percent rpm indicating ranges.

3.12 Voltmeter. An alternating current voltmeter conforming to MIL-M-10304, as applicable, shall be provided on the control panel as shown on Figure 2. The indicating range shall be O to 30V. The design shall be suitable for measuring three phase alternating current voltage generated by two pole and four pole aircraft tachometer generators wherein the frequency may vary over a range of 10 to 70 cycles per second. The scale length shall be a minimum of 2 inches. The case shall be of the 3-1/2 inch size. An accuracy of $\pm$ 2 percent shall apply. The resistance of the meter shall be such that it does not draw sufficient current to unbalance the resistance load during the accomplishment of tachometer generator output voltage tests.

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3.13 Load resistors. Two sets of three wire wound resistors connected in "y" as shown on Figure 3 shall be provided. One set shall incorporate 20 ohm resistors and the other set 40 ohm resistors. The resistance value of each resistor employed shall not differ by more than + 5 percent from the respective value specified. In addition, a set of 15 ohm and a set of 30 ohm load resistors connected in delta shall be provided, with attendant changes in the associated selector switches. THe wattage rating of the resistors shall be 20W.

3.13.1 Mounting and location. The load resistors shall be grouped and secured underneath the panel by means of suitable mounting brackets. The resistors assemblies shall be accessible and conveniently located.

3.13.2 Selector switches. Selector switches as shown on Figure 2 shall be incorporated for facilitating the connector of the load resistor assemblies and the voltmeter to the output of the tachometer generator undergoing test.

3.14 Electrical connectors. Electrical connectors unless otherwise specified shall conform to MIL-C-5015.

3.15 Wiring. The wire employed for the 115V, 400 cycle electrical power circuit shall be suitable for the purpose intended and shall be such that standard electrical ratings are not exceeded. Unless otherwise specified, wire required for the adapter cables specified herein and for connecting the components mounted on the panel shall conform to MIL-W-16878. The wiring shall be grouped together wherever practicable and shall be covered with insulation sleeving conforming to MIL-I-7444. The interconnecting wiring shall be adequately secured within the test set. Hookup wire required for internally connecting the component parts of the rotational speed indicating system shall conform MIL-W-16878.

3.16 Connector receptacles. A connector receptacle conforming to the MS3102A-10SL-3P shall be mounted on the control panel aft and to the left of the tachometer generator mounting pad. It shall be identified as the “test generator input” receptacle. An additional connector receptacle conforming to MS3102A-1OSL-3P and identified as the “generator output” receptacle shall be provided as shown on Figure 2. The “test generator input” receptacle shall be electrically connected so as to permit the output of individual tachometer generators undergoing test to be connected to the resistance load and associated voltage measuring circuits or to the “generator output” receptacle when the AN3226-1 switch is placed in the “test generator” position. The tachometer generator test circuits discussed above shall incorporate, as applicable, a connector scheme of A to A, B to B, and C to C.

3.17 Switches. The rotary-type switches required for the control panel shall be ruggedly constructed. The basic design employed shall incorporate a positive mechanical index locating each contact position. The index mechanism shall be designed to prevent the movable element from coming to rest between contact positions. The electrical contacts shall be self cleaning. The material used for the contacts and the processing of the material shall be suitable for the purpose intended. Stops shall be provided to limit the travel of the switches beyond the design range and shall be sufficiently rugged to prevent damage to the mechan

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3.18 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872. Solder used for the electrical connections shall be in accordance with QQ-S-571, and of a suitable composition.

3.18.1 Soldering flux. Only rosin, rosin and alcohol, or equivalent plastic rosin mixtures, shall be used as a flux in the assembly of the electrical wiring.

3.19 Power line connector cable. A rubber jacketed flexible cable 10 feet + 2 inches in overall length and using 3 stranded conductors capable of adequately carrying the current required by the test set shall be provided. The cable shall be wired in accordance with MIL-STD-1653 for 400 Hz, single phase, 115 volts. One of the conductors shall be used for grounding the test set. The cable shall be equipped on one end with an MS3106A-20-4S connector plug which shall mate with an MS3102R-2O-4P receptacle mounted in a suitable location on the control panel. The other end shall terminate with an MS3507A-20-4P connector plug. The cable shall be stowed in the stowage compartment.

3.20 Adapter cables. Three adapter cables as shown in Figure 4 shall be furnished with the test set for facilitating the connection of aircraft tachometer generators and indicators undergoing tests. These adapters shall be stowed in the stowage compartment. An additional adapter having the same desgin characteristics of adapter B with the exception that one end shall be equipped with an AN3057-6 adapter and an MS3106E-14S-5S connector shall be provided. An additional adapter shall be provided which shall have the same configuration as adapter B, except the overall length shall be 20 ± 5 feet. Two tee adapters shall also be provided so that the digital indicator section of the test set may be connected in parallel with indicators installed in aircraft. One of these tee adapters shall be suitable for use with tachometer indicators designed to mate with an MS3106-1OSL-3S connector plug; the other tee adapter shall be suitable for use with tachometer indicators designed to mate with an MS3106E-14S-5S connector plug.

3.21 Weight. The weight of the test set shall not exceed 40 pounds.

3.22 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.23 Finishes and Protective coatings.

3.23.1 Aluminum alloy parts. Aluminum alloy parts shall be convered by a anodic film conforming to MIL-A-8625. Small holes and inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541. Where the primary purpose of the treatment is to affort a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing, When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

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3.24 Screw threads. Unless otherwise specified, the threads of all machine screws 0,060 inch or larger in diameter shall conform to MIL-S-7742.

3.25 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.26 Workmanship. The test set, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burns, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts.

3.26.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.26.2 Cleaning. The test set shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.27 Operating instructions. A chart of operating instructions shall be permanently attached to the inside surface of the case lid. The background of the chart shall be white and the marking black. The chart shall withstand effects of the test specified herein and shall clearly define the setup, connections and operation of the test.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article Inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall consist of the tests outlined in Test methods (see 4.6).

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

a. Individual tests (see 4.4.1)
b. Sampling tests (see 4.4.2)

4.4.1 Individual tests. Each test shall be subjected to the following tests as described under 4.6 Test methods:

a. Examination of product
b. Operation at room temperature
c. Resistance load - output voltage

4.4.2 Sampling plans and test.

4.4.2,1 Sampling plan A. One test set selected at random from each 50 or less produced on the contract or order shall be subjected to the following test as described under 4.6 Test methods:

a. Individual tests
b. Low temperature
c. High temperature

4.4.2.2 Sampling plan B. Unless otherwise specified, 1 test set selected at random from the first 10 items of the contract or order shall be subjected to the following tests as described under 4.6 Test methods:

a. Sampling plan A
b. Vibration
c. Endurance
d. Humidity

4.4.2.3 Rejection and retest. When one item selected from a production run fails to meet he specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.4.2,3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the acquiring activity of all defects likely to be found and methods of correcting them.

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4.5 Test conditions.

4.5.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of he test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximate y 29.92 i riches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified condition.

4.5.2 Master rotational speed indicating system. A master rotational speed indicating system shall be used for checking the accuracy of the digital speed indicator of the test set. The master indicating system shall be capable of providing measurement accuracies of + 1.0 revolution per minute and + 0.02 percent rpm.

4.5.3 Attitude. Unless otherwise specified, the test set will be tested in the normal operating position.

4.6 Test methods.

4.6.1 Examination of product. Each test set shall be inspected to determine compliance with the requirements specified herein with respect to dimensions, material, workmanship, and marking.

4.6.2 Operation at room temperature. A transducer, capable of providing output signals for the master rotational speed indicating system, shall be attached to the tachometer generator mounting pad on the control panel and electrically connected to the master rotational speed indicating system. The test set shall be connected to a 115V, 400 cycle single phase power source. The test set and the master rotational speed indicating system shall be electrically energized. The variable speed drive control of the test set shall be rotated until the digital indicator of the test set registers approximately 4200 rpm. The test set shall be allowed to warm up under this o crating condition for a period not to exceed 5 minutes. Immediately after the warmup period, the accuracy of the digital speed indicator of the test set shall be checked at the test speeds specified in Table I by means of the master rotational speed indiating system. The readings registered by the digital indicator shall not differ respectively from the readings registered by the master rotational speed indicating system by more than the tolerance specified in Table I. During the accomplishment of this test, it shall be noted that the controls and switches are easy to operate and the variable speed drive is capable of maintaining the output speed at the specified test points sufficiently constant to result in steady indication of the digital speed indicator. Upon completion of the above, a Type E-30 tachometer indicator conforming to MIL-I-25686, or any other tachometer generator approved by the acquiring activity, shall be electrically connected to the “generator output” receptacle by means of adapter “B” shown on Figure 4,

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The output of the AN5547-2 four pole tachometer generator incorporated within the test shall be connected to the Type E-30 tachometer indicator by properly positioning the AN3226-1 switch. The RPM-PERCENT RPM switch shall be placed in the RPM x 2 range position, The test set shall be electrically energized and the variable speed drive control operated to cause the Type E-30 tachometer indicator to register within its complete design range. The test set shall be capable of driving the tachometer indicator undergoing test smoothly over the design range. Pointer oscillation shall not be encountered with the Type E-30 indicator when readings are taken at 1000, 2000, 3000, and 4000 rpm settings.

TABLE I

Digital Speed Indicator

Test Speed Settings		Room Temperature Tolerance			High Temperature Low Temperature After Vibration After Endurance Tolerance			After Humidity Tolerance		
		Switch Positions			Switch Positions			Switch Positions		
Percent	RPM	RPM x 1	RPM x 2	PERCENT RPM	RPM x 1	RPM x 2	PERCENT RPM	RPM x 1	RPM x 2	PERCENT RPM
0	0	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
5	210	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
10	420	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
20	840	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
30	1260	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
40	1680	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
50	2100	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
60	2520	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
70	2940	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
80	3360	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
90	3780	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
100	4200	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
110	4600	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4
119	5000	±10	±20	±0.2	±15	±30	±0.3	±20	±40	±0.4

4.6.3 Resistance load - output voltage. AN MS28054-2 two-pole tachometer generator shall be attached to the mounting pad located on the control panel. The generator shall be electrically connected to the “test generator input” receptacle by means of adapter “A” shown on Figure 4. The test shall be electrically energized and the variable speed drive control rotated until the digital indicator registers approximately 25 percent rpm. The resistance load switch shall be rotated to the 20 ohm position and the voltage checked across the generator ouput terminals A-B, B-C, and A-C, by means of the voltmeter and associated selector switch provided on tne control panel. The voltage

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measurements shall also be checked by means of a master voltmeter having a known accuracy. The readings established by the voltmeter provided on the control panel shall not differ from respective readings established by the master voltmeter by more than the tolerance specified in Table II (room temp). The test shall be repeated with the output speed of the variable drive set to approximately 100 percent rpm as registered by the digital indicator and the resistance load switch rotated to the 40 ohm position. The same tolerance shall apply.

TABLE II

Resistance Load - Output Voltage

Digital Indicator Reading in Percent RPM	Load Switch Position	Generator Receptacle Test Pins	Test Set Voltmeter Tolerance in Volts		
			Room Temp	Low Temperature High Temperaure After Vibration After Endurance	After Humidity
25	20	A to B	±0.45	0.675	0.9
25	20	B to C	±0.45	0.675	0.9
25	20	A to C	±0.45	0.675	0.9
100	40	A to B	±0.45	0.675	0.9
100	40	B to C	±0.45	0.675	0.9
100	40	A to C	±0.45	0.675	0.9

4.6.4 Low temperature. The test set shall be placed in a test chamber and maintained at a temperature of -55°C +2°C for a period of 24 hours. At the end of the 24 hour period, the temperature shall be changed to -40°C +2°C for a period of 4 hours. The test set while at -40°C +20°C shall be subjected again to the tests specified in 4.6.2 and 4.6.3. Applicable tolerances shall be as specified in Tables I and II. No damage shall be noted which would adversely affect subsequent operation or accuracy of the test set.

4.6.5 High temperature. The test set shall be placed in a test chamber and maintained at a temperature of +71°C +2°C for a period of 24 hours. At the end of the 24 hour period the temperature shall be changed to +55°C +2°C for a period of 4 hours. The test set while at +55°C +2°C shall be subjected again to the tests specified in 4.6.2 and 4.6.3. Applicable tolerances shall be as specified in Tables I and II. No damage shall be noted which would adversely affect subsequent operation or accuracy of the test set.

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4.6.6 Vibration. The test set with the lid closed and secured shall be mounted on a vibration test stand. The test set shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 0.03 +0.006 inch. The frequency shall be varied uniformly from 10 to 55 cycles per second and returned to 10 cycles per second in approximately 1 minute. The vibration shall be applied in each of the following directions for a period of 1 hour:

a. Horizontally, parallel to the major horizontal axis of the test set.
b. Horizontally, at right angles to the major horizontal axis of the test set.
c* Vertically.

Upon completion of the vibration test, the test set shall be inspected to determine that no fixed parts have become loose or damaged. The test set shall be subject again to the test specified in 4.6.2 and 4.6.3. Applicable tolerances shall be as specified in Tables I and II.

4.6.7 Endurance. A Type E-30 tachometer indicator shall be electrically connected to the AN5547-2 tachometer generator incorporated within the test set as outlined under 4.6.2. The test set shall be intermittently operated (15 +2 minutes "on" and 15 +2 minutes “off”) for a total period of 48 hours with the variable speed drive control set to cause the Type E-30 tachometer indicator to register 4000 +50 rpm. The test may be conducted on a continuous basis or in 8 hour intervals. Upon completion of the test, the test set shall again be subjected to the tests specified in 4.6.2 and 4.6.3. Applicable tolerances shall be as specified in Tables I and II.

4.6.8 Humidity. The humidity test shall be in accordance with procedure I of MIL-E-5272 During accomplishment of the test, the test set shall be mounted in its normal operating position with the lid open. Within 1 hour after the end of the 240 hour period, the test set shall again be subjected to the tests specified in 4.6.2 and 4.6.3. Applicable tolerances shall be as specified in Tables I and II. No corrosion shall be noted which would adversely affect subsequent operation.

5. PACKAGING

5.1 Application. The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Levels of preservation and packaging.

5.2.1 Level A. Test sets shall be individually preserved and packed in accordance with MIL-P-116, method II.

5.2.2 Level B. Test sets shall be preserved and packaged in accordance with manufacturer’s commercial practice. Unit quantities shall be as specified by the procuring activity. Packaging material in direct contact with unprotected surfaces susceptible to damage by corrosion shall meet MIL-B-121, MIL-P-4185, MIL-A-148, or MIL-B-130.

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5.2.3 Level C. Test sets shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.3 Levels of packing.

5.3.1 Level A. Test sets preserved and packaged to meet 5.2.1 shall be packed in exterior type shipping containers meeting PPP-B-591, PPP-B-621, PPP-B-636, MIL-B-138, PPP-B-585, PPP-B-601, or MIL-B-10377. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet the special requirements table of PPP-B-636, as applicable.

5.3.2 Level B. Test sets preserved and packaged to meet 5.2.1 shall be packed in domestic type exterior containers meeting PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-585, PPP-B-636, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet the special requirements table of PPP-B-636, as applicable.

5.3.3 Level C. Packages which requires over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.4 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100 except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of JAN-P-100 shall be waived when preservation, packaging, and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

5.5 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Test Set, Indicator-Generator, Tachometer TTU-27/E,
Aircraft Instruments, Field

5.5.1 Reinspection date. Reinspection date shall be in accordance with ANA Bulletin 405.

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6. NOTES

6.1 Intended use. The test set covered by this specification is intended to be used as a portable field instrument for checking the performance characteristics of aircraft tachometer indicators and generators. Representative types of equipment which will be checked by the test set are covered by MIL-G-5413, MIL-G-9398, MIL-I-5573, MIL-I-6470, MIL-I-8401, and MIL-I-6329.

6.2 Ordering data. Acquisition documents should specify the following:

a. Level of packaging and packing desired.
b. First article requirements (see 3.1 and 4.3)
c. Selection of applicable levels of preservation and packaging, and packing.

(1) Level B preservation and packing is intended to provide economical but limited protection, and should be specified only when it is determined the test set will be held in covered storage approximately 6 months from date of initial packaging.

6,3 Storage surveillance. Items preserved and packaged in accordance with level B requirements must be inspected to determine condition when not used within the time period indicated. Items not used within the time period specified must either be represerved or repackaged in accordance with level B requirements in this specification or with level A requirements if storage beyond an additional year is anticipated.

6.4 Definition.

6.4.1 Percent RPM. One hundred percent rpm is equivalent to an output speed of 4200 revolutions per minute.

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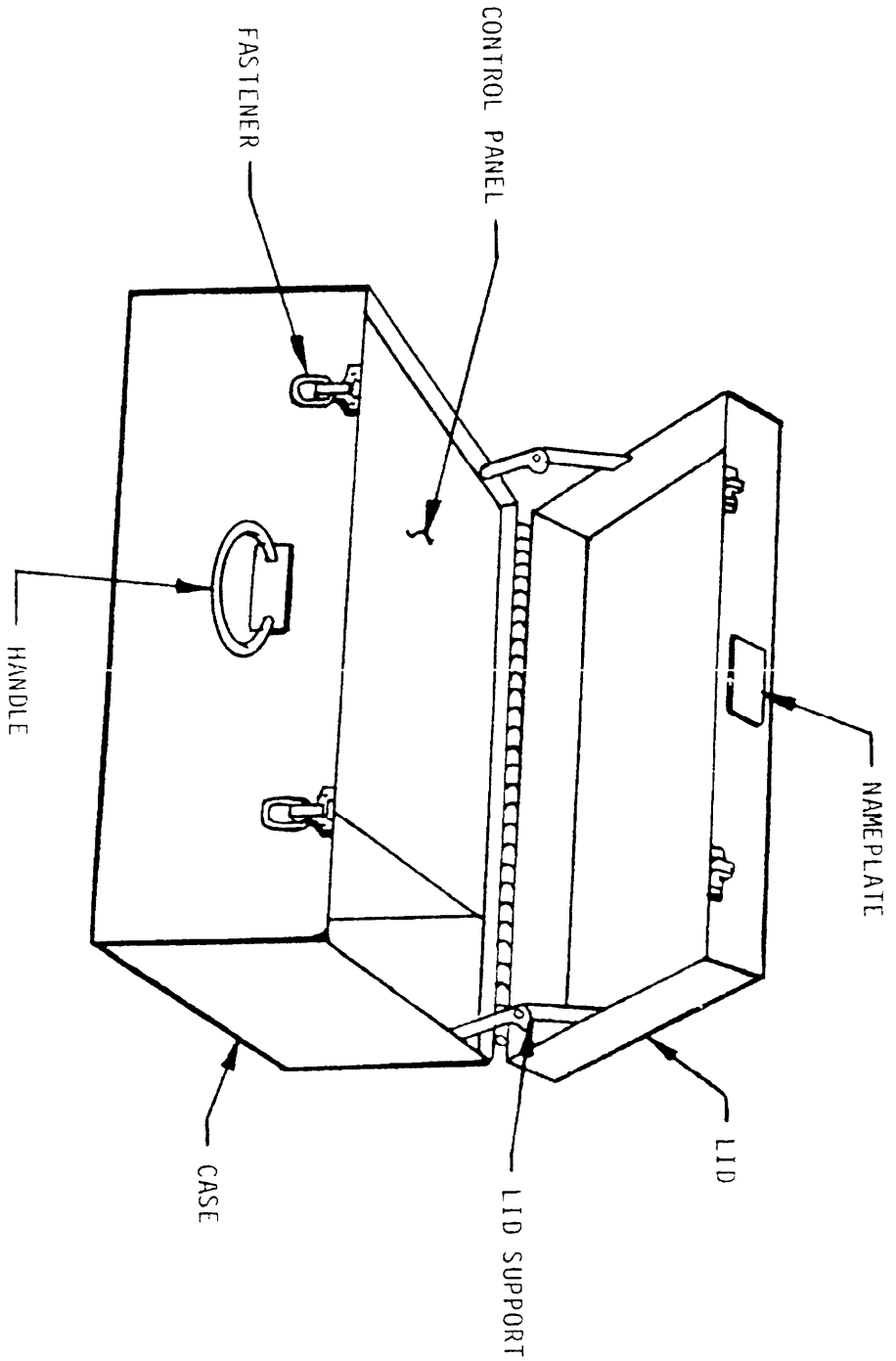


Figure 1 Case (See 3.5.1)

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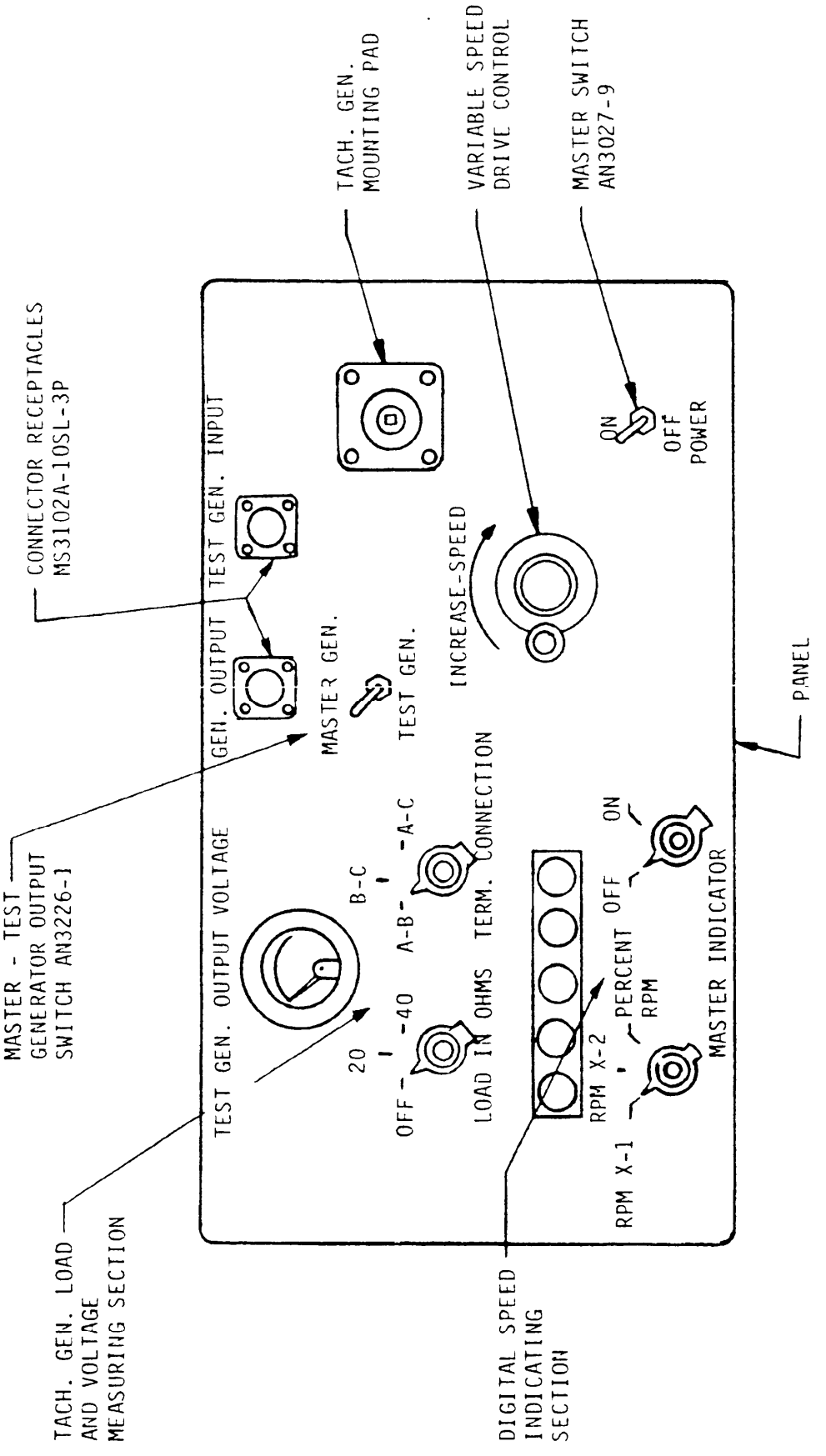
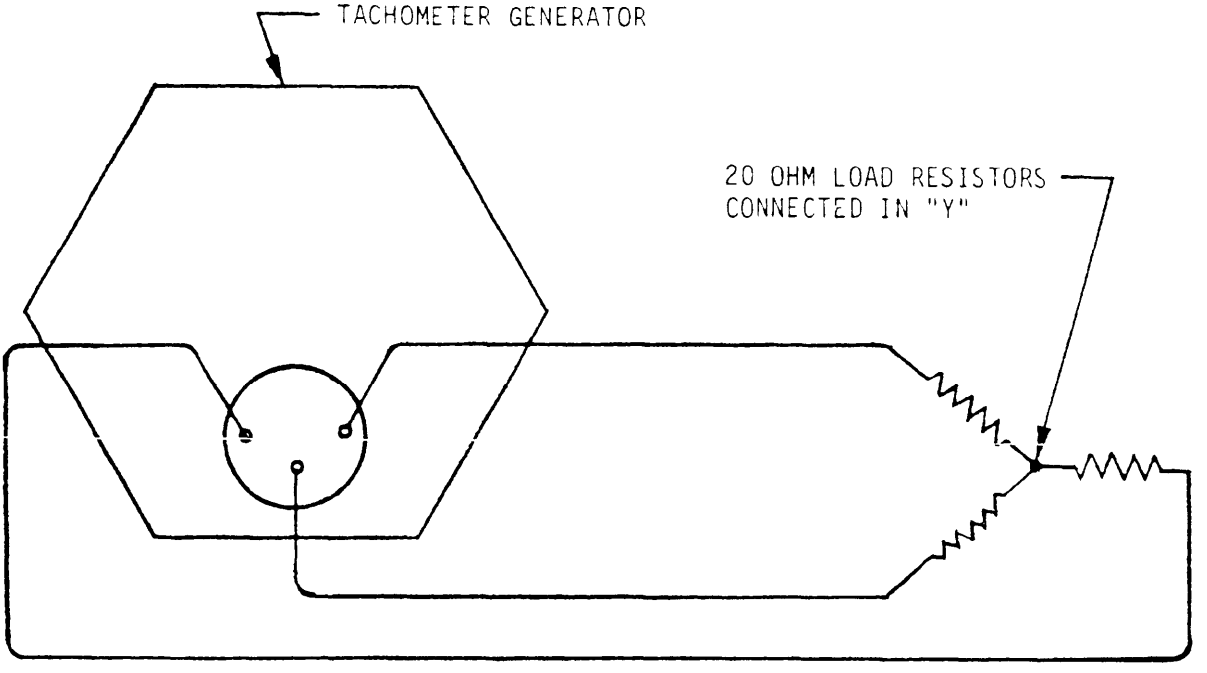


Figure 2. Control panel (See 3.5.1

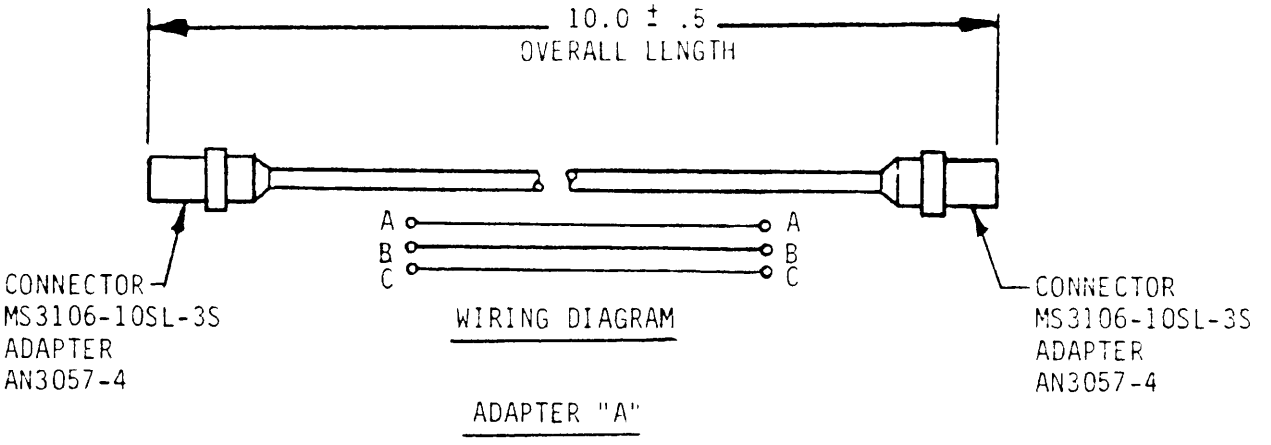
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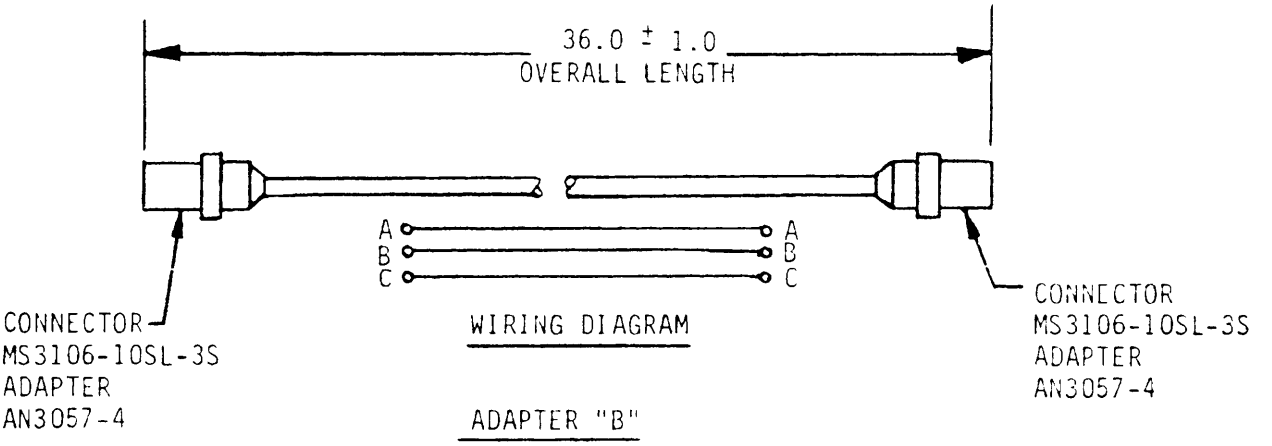
NOTE: THE 40 OHM LOAD RESISTORS SPECIFIED SHALL BE CONNECTED IN THE SAME MANNER SHOWN ABOVE.

Figure 3. Schematic resistor (See 3.13)

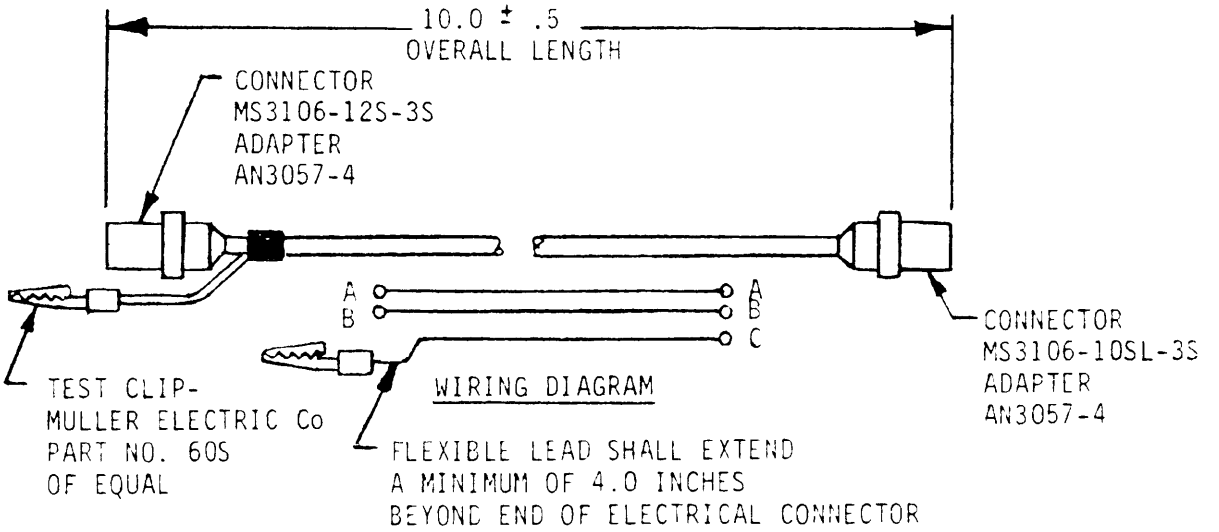
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ADAPTER "A"



ADAPTER "B"



ADAPTER "C"

Figure 4 Adapter Cables (See 3.20)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-26219A	TEST SET, INDICATOR-GENERATOR TTU-27/E ACFRT INSTR. FIELD

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* - Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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